# Compact reciprocal Maxwell scatterers can approach unit absorption asymmetry

Free-space cross sections, radiation-consistent dipoles, and a finite-aperture localization proof

OQ7 research continuation

Mathematical proof, numerical diagnostics, and independent review

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**Abstract**

We study the extinction-normalized absorption difference of opposite time-reversed plane waves incident on a finite compact reciprocal Maxwell scatterer. For every fixed vacuum frequency and every prescribed upper permittivity and loss caps $P > 1$ and $Q > 0$, we prove that the supremum of $\Gamma = |\sigma_{\mathrm{abs},+} - \sigma_{\mathrm{abs},-}|/\sigma_{\mathrm{ext}}$ is one, already within local, nonmagnetic, piecewise homogeneous anisotropic dielectric bodies with $I \preceq \mathrm{Re}_H\, \varepsilon \preceq PI$ and $0 \prec \mathrm{Im}_H\, \varepsilon \preceq QI$. Every body is compact and every coefficient finite; the sequence has no common size or total material-resource bound. The value one is unattainable by any individual regular compact object with positive extinction. The main analytic step proves that cross sections of finite-aperture truncations of a fixed strictly absorbing planar stack, divided by aperture area, converge to the exact planar values. A volume integral equation supplies aperture-independent coercivity, and mixed Fourier estimates control the grazing-radiation singularity. All-angle scattering follows from the exact optical theorem. For square apertures we give a nonoptimal $O((kR)^{-1/3})$ convergence estimate at fixed material parameters. We also derive the sharp complete electric–magnetic dipole cross-section bound, provide exact radiation-damped equality tensors, and test finite objects with a full-vector retarded discrete-dipole solver. Known dipolar and finite-plate precedents are distinguished from the present ordinary mathematical proof. Priority is not certified, and no Lean claim is made in this phase.

## 1 The precise question and the main conclusion

Miller and Monticone's Section 4.1 asks for fundamental bounds on absorption–scattering asymmetry under time-reversed illumination [1]. Extinction equality is established reciprocal scattering theory [2]; separate absorption and scattering equality does not follow. The present addendum answers one precise resource-free plane-wave version of that question. It does not replace the earlier restricted thickness theorem or its hypotheses.

Use the time convention $e^{-i\omega t}$ and a real vacuum wave number $k = \omega/c > 0$. For a unit transverse electric polarization $e$ and a real unit propagation vector $d$, the two incident waves are

$$\begin{aligned} E^{+}_{\mathrm{inc}}(x) &= e\, e^{ikd\cdot x}, & Z_0 H^{+}_{\mathrm{inc}}(x) &= (d\times e)e^{ikd\cdot x},\\ E^{-}_{\mathrm{inc}}(x) &= \bar{e}\, e^{-ikd\cdot x}, & Z_0 H^{-}_{\mathrm{inc}}(x) &= -(d\times \bar{e})e^{-ikd\cdot x}. \end{aligned} \tag{1}$$

They have equal incident intensity, and the magnetic minus sign is essential. For a compact object, each cross section is a total power divided by that incident *intensity*; it has units of area. In particular,

$$\sigma_{\mathrm{ext}} = \sigma_{\mathrm{abs}} + \sigma_{\mathrm{sca}}, \qquad \sigma_{\mathrm{sca}} = \int_{S^2} |F(n)|^2 \,\mathrm{d}\Omega(n), \tag{2}$$

where $E_{\mathrm{sca}}(rn) = e^{ikr}F(n)/r + O(r^{-2})$ for unit electric incident amplitude. This includes every outgoing direction and polarization. It is not reflected power alone or a finite numerical collection cone. Reciprocity gives $\sigma_{\mathrm{ext},+} = \sigma_{\mathrm{ext},-} = E$. When $E > 0$, set

$$\Gamma = \frac{|\sigma_{\mathrm{abs},+} - \sigma_{\mathrm{abs},-}|}{E} = \frac{|\sigma_{\mathrm{sca},+} - \sigma_{\mathrm{sca},-}|}{E}. \tag{3}$$

**Theorem 1.1** (Unrestricted compact dielectric supremum)**.** *Fix $k > 0$, $P > 1$, and $Q > 0$. For every $\epsilon > 0$ there is a finite compact body consisting of two adjacent homogeneous anisotropic dielectric layers with a square transverse aperture, surrounded by vacuum, such that*

$$\mu = I, \qquad \varepsilon^T = \varepsilon, \qquad I \preceq \mathrm{Re}_H\, \varepsilon \preceq PI, \qquad 0 \prec \mathrm{Im}_H\, \varepsilon \preceq QI$$

*inside the body, and opposite plane waves with the same real linear polarization have positive common extinction and $\Gamma > 1 - \epsilon$. Every dimension and material coefficient is finite. Consequently*

$$\sup \Gamma = 1$$

*over this class. For each individual regular compact scatterer with positive extinction and the complete optical theorem, $\Gamma < 1$.*

The upper bound and its strictness are proved below. The constructive lower bound follows from the finite-aperture localization theorem and the exact two-layer planar construction, with an ordered positive-loss regularization. The proof is supplied in full in the subsequent sections. No finite-dimensional matrix is assumed to have a dielectric completion.

**What is and is not fixed.** Frequency and both pointwise upper material caps can be fixed throughout. The theorem does not impose a common diameter, footprint, thickness, volume, integrated loss, or bandwidth. Every finite member has a positive loss floor, but no common positive lower bound is required along the sequence. These quantifiers are part of the conclusion, rather than omitted resources.

**Relation to the preceding study.** The earlier Theorem 6.4 is an exact infinite-planar Maxwell construction. Its finite-power beam extension still used a laterally infinite body. The present proof supplies the missing compact-material, plane-wave cross-section limit. The earlier scalar inverse-square thickness law and its Lean supplement are preserved without modification. They are not used as a formal verification of the present result.

### 1.1 Strictness and a useful resource-dependent inequality

Passivity gives $0 \le \sigma_{\mathrm{abs},\pm} \le E$, so $\Gamma \le 1$. If $\Gamma = 1$, the more-absorbing experiment has absorption $E$ and zero scattering. The far-field amplitude of a regular compact scatterer is continuous on $S^2$, hence its zero squared integral makes it identically zero. Its forward optical theorem,

$$E = \frac{4\pi}{k}\,\mathrm{Im}\{e^\dagger F(d)\}, \tag{4}$$

then gives $E = 0$, a contradiction. Exact equality at positive extinction is therefore impossible even though the supremum is one.

There is a precise familiar obstruction if forward directivity is controlled. For the more-absorbing experiment $h$, define its co-polar forward directivity by

$$D_{\mathrm{co},h} = \frac{4\pi |e_h^\dagger F_h(d_h)|^2}{\sigma_{\mathrm{sca},h}}.$$

At positive extinction its numerator and denominator are nonzero. Squaring (4) and using $|\Im z| \le |z|$ gives

$$\sigma_{\mathrm{sca},h} \ge \frac{k^2 E^2}{4\pi D_{\mathrm{co},h}} = \frac{\pi E^2}{\lambda^2 D_{\mathrm{co},h}}.$$

Since $|\Delta\sigma_{\mathrm{abs}}| = E - \sigma_{\mathrm{sca},h} - \sigma_{\mathrm{abs},\ell}$,

$$\boxed{\Gamma \le 1 - \frac{\pi E}{\lambda^2 D_{\mathrm{co},h}}.} \tag{5}$$

This is a direct consequence of established optical-theorem/directivity tradeoffs [9, 10], not a novelty claim. At a fixed nonzero $E$, approaching one requires unbounded forward directivity. The large-aperture construction permits that growth. No hard multipole cutoff is silently inferred from finite diameter.

## 2 Complete electric–magnetic dipole models

### 2.1 Radiation, reciprocity, and the dimensional normalization

Let $f = (E, Z_0 H)$ and $d_p = (p/\varepsilon_0, Z_0 m)$, and write $d_p = \alpha f$. Define the dimensionless polarizability and its dissipative defect by

$$A = \frac{k^3}{6\pi}\alpha, \qquad Q_A = \frac{A - A^\dagger}{2i} - A^\dagger A. \tag{6}$$

Complete dipolar passivity is $Q_A \succeq 0$ for every incident six-vector. Reciprocity is

$$A^T = DAD, \qquad D = \mathrm{diag}(I_3, -I_3). \tag{7}$$

Thus the electric and magnetic blocks are symmetric and the magnetoelectric blocks have the reciprocal minus-transpose relation.

For incident electric amplitude one, the full dipole cross sections are

$$\begin{aligned} \sigma_{\mathrm{ext}} &= \frac{6\pi}{k^2}\,\mathrm{Im}(f^\dagger A f), \\ \sigma_{\mathrm{sca}} &= \frac{6\pi}{k^2}\,\|Af\|^2, \\ \sigma_{\mathrm{abs}} &= \frac{6\pi}{k^2} f^\dagger Q_A f. \end{aligned} \tag{8}$$

For example, the exact electric far-field amplitude of this model is

$$F(n) = \frac{3}{2k}\left[(I - nn^T)(Af)_E - n \times (Af)_M\right].$$

Integrating its squared norm over the entire sphere gives (8). Electric–magnetic interference integrates to zero in the total radiated power but is retained in its angular distribution. The forward

projection gives the same extinction through (4). These are the established magnetoelectric optical theorem and radiation dressing framework [3]. The full matrix inequality is necessary: an orthogonal eigenpolarizability assumption excludes nonnormal responses relevant here.

Put $J = \mathrm{diag}(I_3, iI_3)$, $\widetilde{A} = J^\dagger A J$, and $S = I + 2i\widetilde{A}$. Then

$$S^T = S, \qquad I - S^\dagger S = 4J^\dagger Q_A J \succeq 0.$$

For the paired plane waves, $f_- = D\bar{f}_+$ and $\|f_+\|^2 = 2$. The vectors $u = J^\dagger f_+/\sqrt{2}$ and $\bar{u}$ are unit and orthogonal:

$$u^T u = \tfrac{1}{2}\{e \cdot e - (d \times e) \cdot (d \times e)\} = 0.$$

Writing $X = 2 - 2\,\mathrm{Re}(u^\dagger S u)$,

$$(\sigma_{\mathrm{ext}}, \sigma_{\mathrm{abs}}, \sigma_{\mathrm{sca}}) = \frac{3\pi}{k^2}(X, 1 - \|Su\|^2, \|(S-I)u\|^2). \tag{9}$$

The area factor is derived from the dipole cross sections. It is not a normalization of an infinite free-space plane wave as a unit finite-power state in the full Maxwell channel space.

**Theorem 2.1** (Sharp complete dipole bound; established passivity consequence)**.** *For a reciprocal passive electric–magnetic dipole model,*

$$0 \le \sigma_{\mathrm{ext}} \le \frac{12\pi}{k^2}, \qquad |\Delta\sigma_{\mathrm{abs}}| \le \sigma_{\mathrm{ext}} - \frac{k^2\sigma_{\mathrm{ext}}^2}{12\pi}.$$

*Every permitted extinction value has an equality model. For positive extinction,*

$$\Gamma \le 1 - \frac{k^2\sigma_{\mathrm{ext}}}{12\pi} < 1, \qquad \sup\Gamma = 1. \tag{10}$$

*Proof.* Reciprocity equates the complex forward overlaps, hence the two extinctions. By Cauchy–Schwarz, $\|Su\|^2 \ge [\mathrm{Re}(u^\dagger Su)]^2 = (1 - X/2)^2$. Both outputs also have norm at most one. Thus the absorption difference is at most $X - X^2/4$, and (9) gives the result. Equality in the interior occurs precisely when, after exchanging the experiments, $Su = (1 - X/2)u$ and $\|S\bar{u}\| = 1$. The next explicit family supplies equality for every $X$; the endpoints follow by continuity. □

The same coefficient has a direct radiation interpretation. The dipole far-field formula and Cauchy–Schwarz give $|e^\dagger F(d)|^2 \le 9\,\|Af\|^2/(2k^2)$, so every such dipole has $D_{\mathrm{co}} \le 3$. Substituting this into (5) recovers (10). This dipolar directivity restriction is not a bound for arbitrary large compact Maxwell objects.

## 2.2 Exact equality tensors and radiation removal

On the active physical coordinates $(E_x, Z_0H_y, E_y)$, choose $-1 < t < 1$ and $b = \sqrt{(1-t^2)/2}$. Set every other six-dimensional entry to zero and put

$$A = \frac{i}{2}\begin{pmatrix} 1-t & 0 & -b \\ 0 & 1-t & -b \\ -b & b & 1 \end{pmatrix}. \tag{11}$$

It obeys the reciprocal signs, and direct multiplication gives

$$Q_A = \frac{1}{4}qq^T, \qquad q = (b, b, -t)^T.$$

For $f_+ = (1,1,0)^T$ and $f_- = (1,-1,0)^T$,

$$\begin{aligned} \sigma_{\mathrm{ext},+} = \sigma_{\mathrm{ext},-} &= \frac{6\pi}{k^2}(1-t), \\ \sigma_{\mathrm{abs},+} &= \frac{3\pi}{k^2}(1-t^2), \quad \sigma_{\mathrm{abs},-} = 0, \\ \sigma_{\mathrm{sca},+} &= \frac{3\pi}{k^2}(1-t)^2, \quad \sigma_{\mathrm{sca},-} = \frac{6\pi}{k^2}(1-t). \end{aligned} \tag{12}$$

Hence $\Gamma = (1+t)/2 \to 1$ as $t \uparrow 1$. In this dipole sequence the total extinction tends to zero.

Radiation reaction has not been dropped. Let $a = \sqrt{2(1-t)/(1+t)}$ and, on the same active subspace, define

$$A_0 = i \begin{pmatrix} a^2 & -a^2/2 & -a \\ a^2/2 & 0 & -a \\ -a & a & 1 \end{pmatrix}.$$

Both relevant inverses exist for $-1 < t < 1$, and

$$A = (A_0^{-1} - iI)^{-1}, \qquad \frac{A_0 - A_0^\dagger}{2i} = vv^T \succeq 0, \quad v = (a,0,-1)^T.$$

Thus (11) is exactly radiation-dressed from a passive bare response, with the inactive subspace treated separately.

An instructive identity for every reciprocal polarizability is

$$\sigma_{\mathrm{abs},+} - \sigma_{\mathrm{abs},-} = \frac{6\pi}{k^2} f_+^\dagger (AA^\dagger - A^\dagger A) f_+. \tag{13}$$

Indeed $\|Af_-\|^2 = f_+^\dagger AA^\dagger f_+$ by (7), while extinction is equal. Every normal reciprocal dipole tensor therefore has zero asymmetry. This is consistent with established nonorthogonal-mode explanations [5].

**Restrictions and realization gaps.** Restricting the response to one electric and one magnetic coordinate gives the earlier scalar two-port feasible set and its established constant $\Gamma \le \sqrt{10\sqrt{5}-22} \simeq 0.6005662120$. A third radiating coordinate removes that restriction. For real linear polarization, a purely electric point response has identical local incident electric fields and zero asymmetry. For conjugate circular polarizations, an electric-only three-coordinate example exists; its exact formulas are retained in the research archive. None of these radiation-consistent point tensors alone proves realization by a local finite dielectric. Inverting a quasistatic sphere polarizability near a resonance leaves dynamic-depolarization and higher-multipole errors to control [15, 16]. The following compact-object proof avoids that completion assumption.

## 3 Finite-aperture localization and compact dielectric realizations

This section passes from an exact planar dielectric construction to ordinary compact three-dimensional Maxwell scatterers. The decisive estimate is a volume-current estimate for each fixed strictly absorbing stack. Scattered power is then obtained from the complete optical theorem, so all radiation directions, polarization conversion, and the forward diffraction peak remain included. The argument is analytical; no finite-dimensional matrix completion or numerical convergence assumption enters it. No claim of priority is made.

### 3.1 Geometry, conventions, and the localization statement

Fix a real vacuum wave number $k > 0$ and the time convention $e^{-i\omega t}$. Write a spatial point as $\boldsymbol{r} = (x, z) \in \mathbb{R}^2 \times \mathbb{R}$. Let $\Omega \subset \mathbb{R}^2$ be bounded, open, and Lipschitz, with $|\Omega| > 0$, and let $L > 0$. Suppose that the measurable electric susceptibility $\chi : (0, L) \to \mathbb{C}^{3\times 3}$ satisfies

$$\chi(z)^T = \chi(z), \qquad \operatorname{Im}\chi(z) \succeq \delta I_3, \qquad \|\chi(z)\| \le K \quad \text{for almost every } z, \tag{14}$$

where $K < \infty$ and $\delta > 0$ are fixed. For matrices, $\operatorname{Im} B = (B - B^\dagger)/(2i)$ and $\operatorname{Re} B = (B + B^\dagger)/2$; inequalities between Hermitian matrices are in the quadratic-form sense.

For $R > 0$, set

$$D_R = (R\Omega) \times (0, L), \qquad \mu_R = I_3, \qquad \epsilon_R(\boldsymbol{r}) = \begin{cases} I_3 + \chi(z), & \boldsymbol{r} \in D_R, \\ I_3, & \boldsymbol{r} \notin \overline{D_R}. \end{cases} \tag{15}$$

Values on the boundary do not affect the weak problem. Each material support is compact. The aperture is sharp; smooth windows introduced below modify only a trial polarization, never the material or incident field.

Let $e \in \mathbb{C}^3$ satisfy $e \cdot e_z = 0$ and $|e| = 1$. The two incident electric fields are

$$f_+(z) = e\, e^{ikz}, \qquad f_-(z) = \overline{e}\, e^{-ikz} = \overline{f_+(z)}. \tag{16}$$

These are time-reversed incident vacuum waves. Their corresponding magnetic fields transform with the usual additional minus sign. Cross sections are divided by the incident intensity; with electric amplitude one this gives the formulas below without an impedance factor. The planar scattered power is always defined relative to propagation through vacuum over the same distance.

**Theorem 3.1** (Localization of exact Maxwell cross sections)**.** *Under (14)–(16), let $\sigma^{\pm}_{\mathrm{ext},R}$, $\sigma^{\pm}_{\mathrm{abs},R}$, and $\sigma^{\pm}_{\mathrm{sca},R}$ be the exact plane-wave cross sections of (15). Let $X^{\pm}_\infty$, $A^{\pm}_\infty$, and $C^{\pm}_\infty$ be the normal-incidence extinction, absorption, and scattered-power densities of the infinite planar stack with the same $\chi$ and $L$. Then*

$$\frac{\sigma^{\pm}_{\mathrm{ext},R}}{R^2|\Omega|} \longrightarrow X^{\pm}_\infty, \qquad \frac{\sigma^{\pm}_{\mathrm{abs},R}}{R^2|\Omega|} \longrightarrow A^{\pm}_\infty, \qquad \frac{\sigma^{\pm}_{\mathrm{sca},R}}{R^2|\Omega|} \longrightarrow C^{\pm}_\infty. \tag{17}$$

*The scattering limit includes the entire sphere of outgoing directions and both transverse polarizations. All constants may depend on the fixed stack, including $L$, $K$, and $\delta$; uniformity as $\delta \downarrow 0$ is not asserted.*

We prove this theorem through the current estimate (52). A separate square-aperture estimate will give an $O((kR)^{-1/3})$ upper error bound at fixed material and thickness.

### 3.2 The volume equation and the complete optical theorem

All Hilbert-space inner products are conjugate-linear in their first argument. Extend a polarization $\boldsymbol{P} \in L^2(D_R; \mathbb{C}^3)$ by zero to $\mathbb{R}^3$, and define

$$g_k(\boldsymbol{r}) = \frac{e^{ik|\boldsymbol{r}|}}{4\pi|\boldsymbol{r}|}, \qquad T\boldsymbol{P} = (k^2 I_3 + \nabla \operatorname{div})(g_k * \boldsymbol{P}), \qquad T_R\boldsymbol{P} = (T\boldsymbol{P})|_{D_R}. \tag{18}$$

The derivatives in this formula are distributional. Here $\boldsymbol{P} = \chi E$ is the electric polarization divided by $\epsilon_0$, not the physical electric current $-i\omega\epsilon_0\boldsymbol{P}$. The standard electric volume-integral formulation [11, 12] gives the exact equation

$$\mathcal{L}_R\boldsymbol{P}_R = f|_{D_R}, \qquad \mathcal{L}_R = \chi^{-1} - T_R, \qquad f = f_+ \text{ or } f_-. \tag{19}$$

**Lemma 3.2** (Bounded Green operator and all-angle radiation identity)**.** *For every fixed finite $R$, $T_R$ is bounded on $L^2(D_R;\mathbb{C}^3)$. For every polarization in this space,*

$$\operatorname{Im}\langle \boldsymbol{P}, T_R\boldsymbol{P}\rangle = \frac{k^3}{16\pi^2}\int_{\mathbb{S}^2}\left|(I_3-\widehat{n}\widehat{n}^T)\int_{D_R}\boldsymbol{P}(y)e^{-ik\widehat{n}\cdot y}\,dy\right|^2 d\widehat{n}\geq 0. \tag{20}$$

*The continuous electric far-field amplitude is*

$$F_{\boldsymbol{P}}(\widehat{n}) = \frac{k^2}{4\pi}(I_3-\widehat{n}\widehat{n}^T)\int_{D_R}\boldsymbol{P}(y)e^{-ik\widehat{n}\cdot y}\,dy. \tag{21}$$

*Proof.* Put $g_0(\boldsymbol{r}) = 1/(4\pi|\boldsymbol{r}|)$. The distributional operator $\nabla\operatorname{div}(g_0 * \cdot)$ has Fourier multiplier $-\xi\xi^T/|\xi|^2$, so it is bounded on all of $L^2(\mathbb{R}^3;\mathbb{C}^3)$. This multiplier includes its local distributional term. Near the origin,

$$g_k - g_0 = \frac{ik}{4\pi} + O(|\boldsymbol{r}|),$$

and the second derivatives of this difference are $O(|\boldsymbol{r}|^{-1})$. The kernels of $k^2 g_k I_3$ and $\nabla\operatorname{div}(g_k-g_0)$ are therefore locally integrable weakly singular kernels. On the bounded difference set $D_R - D_R$, their absolute row and column integrals are finite. Schur's estimate, together with the Newtonian Fourier-multiplier bound, proves boundedness. Its constant need not be uniform in $R$.

The smooth identity

$$\operatorname{Im} g_k(\boldsymbol{r}) = \frac{k}{16\pi^2}\int_{\mathbb{S}^2} e^{ik\widehat{n}\cdot\boldsymbol{r}}\,d\widehat{n} \tag{22}$$

follows by integrating the exponential on the sphere. The Green tensor is reciprocal, so its Hermitian imaginary kernel is obtained by applying $k^2 I_3 + \nabla\operatorname{div}$ to $\operatorname{Im} g_k$. Differentiating the smooth angular integral in (22) supplies the factor $k^2(I_3-\widehat{n}\widehat{n}^T)$. Fubini's theorem gives (20), first for smooth polarizations and then by $L^2(D_R)$ density. The real local distributional term makes no contribution to this imaginary quadratic form.

Since $D_R$ has finite measure, every such polarization is in $L^1$. The outgoing large-distance expansion of (18) gives (21); the same $L^1$ fact gives continuity in $\widehat{n}$. The transverse projector and the full sphere in these formulas include both physical polarizations and every radiative direction. □

**Proposition 3.3** (Uniform solvability, energy balance, and reciprocity)**.** *Equation (19) has a unique solution, and*

$$\|\mathcal{L}_R^{-1}\|\leq c^{-1}, \qquad c=\frac{\delta}{K^2}, \tag{23}$$

*independently of $R$. Define*

$$W = -\operatorname{Im}\chi^{-1} = \chi^{-\dagger}(\operatorname{Im}\chi)\chi^{-1}\succeq cI_3. \tag{24}$$

*The exact cross sections satisfy*

$$\sigma_{\mathrm{ext},R} = k\,\operatorname{Im}\langle f, \boldsymbol{P}_R\rangle, \qquad \sigma_{\mathrm{abs},R} = k\,\langle \boldsymbol{P}_R, W\boldsymbol{P}_R\rangle, \tag{25}$$

$$\sigma_{\mathrm{sca},R} = k\,\operatorname{Im}\langle \boldsymbol{P}_R, T_R\boldsymbol{P}_R\rangle = \int_{\mathbb{S}^2}|F_{\boldsymbol{P}_R}(\widehat{n})|^2\,d\widehat{n}, \qquad \sigma_{\mathrm{ext},R} = \sigma_{\mathrm{abs},R}+\sigma_{\mathrm{sca},R}. \tag{26}$$

*For the paired illuminations (16),*

$$\sigma^+_{\mathrm{ext},R} = \sigma^-_{\mathrm{ext},R} > 0. \tag{27}$$

*Proof.* For $v \in \mathbb{C}^3$, $\delta|v|^2 \le \mathrm{Im}(v^\dagger \chi v) \le |v|\,|\chi v|$. Thus $\chi$ is invertible and $\|\chi^{-1}\| \le \delta^{-1}$. Multiplication of $\chi - \chi^\dagger$ by the two inverses proves the factorization in (24). In addition, $|\chi^{-1}v| \ge |v|/K$, which proves its lower bound. In particular $W$ is bounded.

By Lemma 3.2,

$$-\mathrm{Im}\langle \boldsymbol{P}, \mathcal{L}_R \boldsymbol{P}\rangle = \langle \boldsymbol{P}, W\boldsymbol{P}\rangle + \mathrm{Im}\langle \boldsymbol{P}, T_R \boldsymbol{P}\rangle \ge c\|\boldsymbol{P}\|_2^2. \tag{28}$$

The bounded sesquilinear form $a(Q,P) = i\langle Q, \mathcal{L}_R P\rangle$ has $\mathrm{Re}\, a(P,P) \ge c\|P\|_2^2$. The complex Lax–Milgram theorem gives existence and uniqueness for every $L^2$ right-hand side. Applying Cauchy–Schwarz to (28) yields $c\|P\|_2 \le \|\mathcal{L}_R P\|_2$, proving (23). Only coercivity, whose constant is independent of $R$, enters this inverse estimate.

The field $E = f + T\boldsymbol{P}_R$ obeys $E = \chi^{-1}\boldsymbol{P}_R$ in $D_R$ and

$$\nabla \times \nabla \times E - k^2 E = k^2 \boldsymbol{P}_R$$

distributionally. It is therefore the outgoing Maxwell solution with the coefficient (15). To make its regularity explicit, $g_k * \boldsymbol{P}_R$ is locally $H^2$ by the scalar Helmholtz equation and local elliptic regularity. Consequently $E \in L^2_{\mathrm{loc}}$ and $\nabla \times E = k^2 \nabla \times (g_k * \boldsymbol{P}_R) + \nabla \times f \in H^1_{\mathrm{loc}}$. The $H(\mathrm{curl})$ weak formulation and the interface conditions are well defined even at aperture edges and corners. Conversely, an outgoing solution has this Green representation, so the volume-equation uniqueness also gives Maxwell uniqueness.

The absorption density is $kE^\dagger(\mathrm{Im}\,\chi)E = k\boldsymbol{P}_R^\dagger W \boldsymbol{P}_R$. The far-field flux is the sphere integral in (26), and (20) proves its volume expression. Taking the imaginary part of $\langle \boldsymbol{P}_R, \mathcal{L}_R \boldsymbol{P}_R\rangle = \langle \boldsymbol{P}_R, f\rangle = \overline{\langle f, \boldsymbol{P}_R\rangle}$ gives the displayed energy balance with the stated extinction sign. For a plane wave $f(\boldsymbol{r}) = e\, e^{ikd\cdot \boldsymbol{r}}$, (21) also gives the usual forward optical theorem

$$\sigma_{\mathrm{ext},R} = \frac{4\pi}{k}\,\mathrm{Im}(e^\dagger F_{\boldsymbol{P}_R}(d)). \tag{29}$$

Finally, define the complex bilinear pairing $[u,v] = \int_{D_R} u^T v$, without conjugation. The symmetry of $\chi^{-1}$ and the distributional reciprocity of the Green kernel imply $[u, \mathcal{L}_R v] = [\mathcal{L}_R u, v]$. It follows, by substituting inverse images, that $[u, \mathcal{L}_R^{-1} v] = [v, \mathcal{L}_R^{-1} u]$. Since $f_- = \overline{f_+}$,

$$\begin{aligned}\langle f_+, \mathcal{L}_R^{-1} f_+\rangle &= [f_-, \mathcal{L}_R^{-1} f_+] \\ &= [f_+, \mathcal{L}_R^{-1} f_-] = \langle f_-, \mathcal{L}_R^{-1} f_-\rangle.\end{aligned}$$

This proves exact equality of extinction. Neither current is zero, since its right-hand side is nonzero, and $W \succeq cI_3$; hence both absorptions, and therefore their common extinction, are positive. □

### 3.3 The planar problem and the outgoing Weyl fibers

Let $H = L^2((0,L);\mathbb{C}^3)$. For the transverse Fourier transform use

$$\widehat{u}(\kappa) = \int_{\mathbb{R}^2} e^{-i\kappa\cdot x} u(x)\,dx, \qquad u(x) = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} e^{i\kappa\cdot x}\widehat{u}(\kappa)\,d\kappa.$$

Set

$$\beta(\kappa) = \sqrt{k^2 - |\kappa|^2},$$

with positive real value below $|\kappa| = k$ and positive imaginary part above it. Values on the light circle itself will be irrelevant to integration. The outgoing scalar Weyl kernel is

$$G_\kappa(s) = \frac{i}{2\beta(\kappa)} e^{i\beta(\kappa)|s|}. \tag{30}$$

It can be derived by Fourier transforming $(-\Delta - k^2)g_k = \delta$ in $x$, and imposing outgoing or decaying solutions on the two half-lines in $s$.

**Lemma 3.4** (Fiber formula and estimates)**.** *For almost every $\kappa$, the electric Green fiber on $H$ has blocks, with $a, b \in \{1, 2\}$,*

$$(T_\kappa)_{ab} = (k^2\delta_{ab} - \kappa_a\kappa_b)\, G_\kappa *, \tag{31}$$

$$(T_\kappa)_{a3} = (T_\kappa)_{3a} = i\kappa_a\, \partial_z G_\kappa *, \qquad (T_\kappa)_{33} = |\kappa|^2 G_\kappa * - I. \tag{32}$$

*Convolutions in these formulas integrate $z'$ over $(0, L)$. There are finite constants depending only on $k, L$ such that*

$$\|T_\kappa\|_{H\to H} \le C\left(1 + |\kappa| + \frac{k^2 + |\kappa|^2}{|\beta(\kappa)|}\right), \tag{33}$$

$$\|T_\kappa - T_0\|_{H\to H} \le C_0|\kappa|, \qquad |\kappa| \le k/2. \tag{34}$$

*Proof.* The first derivative of $G_\kappa$ away from zero is $-\frac{1}{2}\operatorname{sgn}(s)e^{i\beta|s|}$. Its jump is $-1$, so

$$\partial_z^2 G_\kappa = -\beta(\kappa)^2 G_\kappa - \delta_0. \tag{35}$$

Fourier transforming the transverse derivatives in (18) and using $k^2 - \beta^2 = |\kappa|^2$ proves (31)–(32). In particular the longitudinal local term is exactly $-I$; it is not discarded or replaced by a pointwise Hessian.

Because $\operatorname{Im}\beta \ge 0$, $|G_\kappa(s)| \le (2|\beta|)^{-1}$ and $|\partial_z G_\kappa(s)| \le 1/2$ off the diagonal. Schur's estimate on an interval of length $L$, including the explicit local term, gives (33). For $|\kappa| \le k/2$, $\beta$ stays away from zero and all the operator-valued kernels are differentiable with bounded derivatives on the finite interval. The mean-value estimate gives (34). □

The bound (33) has at most linear growth at large $|\kappa|$. Near the light circle its possible singularity is of order $\big||\kappa| - k\big|^{-1/2}$, which is locally integrable in two transverse dimensions. Its square is not generally locally integrable. This is why a global transverse Plancherel estimate across the light circle is not used.

**Proposition 3.5** (The exact planar solution and its power densities)**.** *For either $f = f_\pm$, the equation*

$$(\chi^{-1} - T_0)q = f \quad \text{in } H \tag{36}$$

*has a unique solution satisfying $\|q\|_H \le c^{-1}\|f\|_H$. Its exact planar power densities are*

$$X_\infty = k\,\operatorname{Im}\langle f, q\rangle_H, \qquad A_\infty = k\langle q, Wq\rangle_H, \qquad C_\infty = k\,\operatorname{Im}\langle q, T_0 q\rangle_H, \qquad X_\infty = A_\infty + C_\infty. \tag{37}$$

*Proof.* At $\kappa = 0$, the transverse kernel of $T_0$ is $(ik/2)e^{ik|z-z'|}I_2$, the mixed entries vanish, and the longitudinal component is the real multiplication operator $-I$. The Hermitian imaginary transverse kernel is precisely

$$\frac{k}{2}\cos(k(z - z'))I_2.$$

Consequently, writing $q_\perp = (q_1, q_2)$,

$$\operatorname{Im}\langle q, T_0 q\rangle_H = \frac{k}{4}\left(\left|\int_0^L e^{ikz} q_\perp(z)\, dz\right|^2 + \left|\int_0^L e^{-ikz} q_\perp(z)\, dz\right|^2\right) \ge 0. \tag{38}$$

The longitudinal term contributes $-\int_0^L |q_3|^2\,dz$, which is real. Thus exactly the coercivity and Lax–Milgram argument of Proposition 3.3 applies to $\chi^{-1} - T_0$ on $H$. This establishes planar well-posedness without treating an infinite plane wave as an $L^2(\mathbb{R}^3)$ incident field.

Extend $q$ by zero outside the interval and reconstruct the planar field by the one-dimensional Green formula. The transverse scattered waves outside the stack have amplitudes

$$b_+ = \frac{ik}{2}\int_0^L e^{-ikz} q_\perp(z)\,dz, \qquad b_- = \frac{ik}{2}\int_0^L e^{ikz} q_\perp(z)\,dz \tag{39}$$

in the $+z$ and $-z$ directions, respectively. Hence $C_\infty = |b_+|^2 + |b_-|^2$, in agreement with (38). For $f_+$, the forward total amplitude in the global vacuum reference is $e + b_+$, so its co-polarized transmission is

$$t_{\mathrm{co}} = 1 + \frac{ik}{2}\int_0^L e^{-ikz} e^\dagger q(z)\,dz, \qquad 2(1 - \operatorname{Re} t_{\mathrm{co}}) = k\,\operatorname{Im}\langle f_+, q\rangle_H. \tag{40}$$

For $f_-$, the same statement uses $\overline{e} + b_-$ and the opposite propagation direction. A boundary-plane transmission coefficient must be divided by $e^{ikL}$ to obtain this reference.

Inside the stack, $E = \chi^{-1} q$; thus $\langle q, Wq\rangle_H = \int_0^L E^\dagger (\operatorname{Im}\chi) E\,dz$. This is the one-dimensional Poynting-flux loss divided by the incident intensity. Taking the imaginary part of (36) gives $X_\infty = A_\infty + C_\infty$, proving (37) with the physical reference phase and both output polarizations. □

### 3.4 Mixed Fourier estimates and the localization proof

**Lemma 3.6** (Smooth-window commutator)**.** *Fix $g \in C_c^\infty(\Omega)$, put $g_R(x) = g(x/R)$, and let $q \in H$. Define, on $\mathbb{R}^2 \times (0, L)$,*

$$H_R = T(g_R q) - g_R T_0 q. \tag{41}$$

*Its transverse Fourier representation is*

$$\widehat{H}_R(\kappa) = \widehat{g}_R(\kappa)(T_\kappa - T_0)q, \qquad \widehat{g}_R(\kappa) = R^2 \widehat{g}(R\kappa). \tag{42}$$

*For each fixed $g$, as $R \to \infty$,*

$$\|H_R\|_{L^2(D_R)} = O_g(1) = o(R). \tag{43}$$

*Proof.* First take $q \in C_c^\infty((0, L); \mathbb{C}^3)$, extend it by zero, and Fourier transform (18) in the transverse variables as a tempered-distribution identity. Lemma 3.4 gives (42). Split its inverse transform into $|\kappa| \le k/2$ and $|\kappa| > k/2$. For the low-frequency part, Hilbert-valued Plancherel and (34) give

$$\|H_R^{\mathrm{low}}\|_{L^2(\mathbb{R}^2; H)} \le C_0 \|q\|_H \|\nabla g_R\|_{L^2(\mathbb{R}^2)} = C_0 \|q\|_H \|\nabla g\|_{L^2(\mathbb{R}^2)}. \tag{44}$$

The equality of the gradient norms uses exactly two transverse dimensions.

For the high-frequency part, fix an integer $N > 3$. Schwartz decay gives, for $R \ge 1$ in fixed length units,

$$|\widehat{g}_R(\kappa)| \le C_{g,N} R^{2-N} |\kappa|^{-N}, \qquad |\kappa| \ge k/2. \tag{45}$$

The fiber bound implies

$$I_N := \int_{|\kappa| \ge k/2} |\kappa|^{-N} (\|T_\kappa\| + \|T_0\|)\,d\kappa < \infty. \tag{46}$$

Indeed, the light-circle singularity is locally $L^1$. At infinity the operator bound is $O(|\kappa|)$, and including the two-dimensional radial measure the integrand is $O(\rho^{2-N})\,d\rho$, where $\rho = |\kappa|$. This

radial upper bound is integrable when $N > 3$, as assumed. The inverse Fourier $L^1$-to-$L^\infty$ estimate, rather than Plancherel, now gives

$$\sup_x \|H_R^{\mathrm{high}}(x,\cdot)\|_H \le C'_{g,N} R^{2-N} \|q\|_H, \tag{47}$$

$$\|H_R^{\mathrm{high}}\|_{L^2(D_R)} \le C''_{g,N} R^{3-N} \|q\|_H. \tag{48}$$

The latter inequality uses $|R\Omega|^{1/2} = R|\Omega|^{1/2}$. Equations (44) and (48) prove (43).

For completeness, the same formulas hold for every $q \in H$. Choose $q_n \in C_c^\infty((0,L);\mathbb{C}^3)$ with $q_n \to q$ in $H$. On every bounded transverse region, boundedness of the restricted three-dimensional Green operator gives $T(g_R q_n) \to T(g_R q)$ in local $L^2$. The low-frequency inverse transform converges in $L^2(\mathbb{R}^2;H)$ by (44), while the high-frequency inverse transform converges in $L^\infty(\mathbb{R}^2;H)$ by (46) and Schwartz decay. Also $g_R T_0 q_n \to g_R T_0 q$ in local $L^2$. Thus the inverse Fourier representation converges to the same outgoing Green field as in the volume equation, proving (42) and its bounds for $q \in H$. No separate grazing-angle regularization is introduced. □

*Proof of Theorem 3.1.* Work with either sign, fixed once throughout the argument. Let $q$ be the solution of (36) and $\boldsymbol{P}_R$ the exact current of (19). For $g \in C_c^\infty(\Omega)$, the trial current $g_R q$ lies in $L^2(D_R)$ and satisfies the exact residual identity

$$\mathcal{L}_R(g_R q) - f = (g_R - 1)f - H_R \quad \text{in } D_R. \tag{49}$$

Applying (23) and then comparing $g_R q$ with the transversely constant $q$ inside $D_R$ gives

$$\|\boldsymbol{P}_R - q\|_{L^2(D_R)} \le R(c^{-1}\|f\|_H + \|q\|_H)\|1-g\|_{L^2(\Omega)} + c^{-1}\|H_R\|_{L^2(D_R)}. \tag{50}$$

For each fixed $g$, divide by $R$ and use Lemma 3.6:

$$\limsup_{R\to\infty} \frac{\|\boldsymbol{P}_R - q\|_{L^2(D_R)}}{R} \le (c^{-1}\|f\|_H + \|q\|_H)\|1-g\|_{L^2(\Omega)}. \tag{51}$$

The left-hand side does not depend on $g$. Density of $C_c^\infty(\Omega)$ in $L^2(\Omega)$ therefore proves

$$\boxed{\|\boldsymbol{P}_R - q\|_{L^2(D_R)} = o(R).} \tag{52}$$

The order of limits is essential: fix the material and a smooth $g$, take $R \to \infty$, and only then approximate 1 in $L^2(\Omega)$. No uniform derivative bound for a sequence of increasingly sharp windows has been assumed.

The incident norm is $\|f\|_{L^2(D_R)} = R|\Omega|^{1/2}\|f\|_H$, and $\|q\|_{L^2(D_R)} = R|\Omega|^{1/2}\|q\|_H$. Moreover (23) gives $\|\boldsymbol{P}_R\|_{L^2(D_R)} = O(R)$. Consequently Cauchy–Schwarz yields

$$\left|\sigma_{\mathrm{ext},R} - kR^2|\Omega|\,\mathrm{Im}\langle f, q\rangle_H\right| \le k\|f\|_{L^2(D_R)}\|\boldsymbol{P}_R - q\|_{L^2(D_R)} = o(R^2). \tag{53}$$

Because $W$ is bounded,

$$\begin{aligned}\left|\langle \boldsymbol{P}_R, W\boldsymbol{P}_R\rangle_{D_R} - R^2|\Omega|\langle q, Wq\rangle_H\right| \\ \le \|W\|_\infty \|\boldsymbol{P}_R - q\|_{L^2(D_R)}(\|\boldsymbol{P}_R\|_{L^2(D_R)} + \|q\|_{L^2(D_R)}) = o(R^2).\end{aligned} \tag{54}$$

Equations (53), (54), and Proposition 3.5 prove the first two limits. Subtract the exact finite-body identity $\sigma_{\mathrm{sca},R} = \sigma_{\mathrm{ext},R} - \sigma_{\mathrm{abs},R}$ and the exact planar identity $C_\infty = X_\infty - A_\infty$ to obtain the third.

In particular, the scattering conclusion does not require convergence of the far field pointwise or uniformly in direction. It follows from its full integrated squared norm in (26), and thus retains a narrowing forward peak and any edge or grazing radiation. □

### 3.5 A quantitative rate for a square aperture

The following square has *side length* $R$, so its area is $R^2$. This convention avoids confusing a side length with a half-width.

**Lemma 3.7** (An interior square cutoff)**.** *Let $S_R = (-R/2, R/2)^2$. There are smooth functions $g_{R,w} \in C_c^\infty(S_R)$, $0 \le g_{R,w} \le 1$, defined for $0 < w \le R/8$, and constants independent of $R, w$, such that*

$$\|1 - g_{R,w}\|_{L^2(S_R)} \le C(Rw)^{1/2}, \qquad \|\nabla g_{R,w}\|_{L^2(\mathbb{R}^2)} \le C(R/w)^{1/2}, \tag{55}$$

$$\|\partial^\alpha g_{R,w}\|_{L^1(\mathbb{R}^2)} \le C_\alpha R w^{-4} \quad (|\alpha| = 5), \qquad |\widehat{g}_{R,w}(\kappa)| \le C R w^{-4} |\kappa|^{-5} \quad (\kappa \ne 0). \tag{56}$$

*Proof.* Fix a smooth $a : \mathbb{R} \to [0,1]$ with $a(u) = 0$ for $u \le 0$ and $a(u) = 1$ for $u \ge 1$. Set

$$a_{R,w}(s) = a\left(\frac{R/2 - w/4 - |s|}{w}\right), \qquad g_{R,w}(x_1, x_2) = a_{R,w}(x_1) a_{R,w}(x_2).$$

The absolute-value cusp is harmless because $a_{R,w}$ is identically one near the origin. Its support lies a distance $w/4$ inside the two endpoints, and it equals one when $|s| \le R/2 - 5w/4$. Hence $g_{R,w}$ is genuinely compactly supported inside $S_R$, and the area where it differs from one is $O(Rw)$. This gives the first estimate in (55).

For every integer $n \ge 1$,

$$\|a_{R,w}^{(n)}\|_{L^1(\mathbb{R})} \le C_n w^{1-n}, \quad \|a'_{R,w}\|_{L^2(\mathbb{R})}^2 \le C w^{-1}, \quad \|a_{R,w}\|_{L^p(\mathbb{R})}^p \le R \quad (p = 1, 2).$$

The product rule and these one-dimensional estimates give the gradient bound. A pure fifth derivative has $L^1$ norm at most $CRw^{-4}$. For a mixed derivative $\alpha = (\alpha_1, \alpha_2)$ with both entries positive and total order five, its norm is at most $Cw^{1-\alpha_1} w^{1-\alpha_2} = Cw^{-3} \le CRw^{-4}$. Thus the derivative estimate holds for every $|\alpha| = 5$. For any $\kappa \ne 0$, select a coordinate $j$ with $|\kappa_j| \ge |\kappa|/\sqrt{2}$. Integrating by parts five times in that coordinate gives

$$|\widehat{g}_{R,w}(\kappa)| \le |\kappa_j|^{-5} \|\partial_j^5 g_{R,w}\|_1 \le CRw^{-4}|\kappa|^{-5},$$

which proves the last estimate. □

**Proposition 3.8** (Square-aperture error bound)**.** *Fix the stack in (14), its thickness $L$, and one of the incidences (16). For $D_R = S_R \times (0, L)$, let $\boldsymbol{P}_R$ and $q$ denote the exact compact and planar currents. There are finite constants independent of $R$ such that, for sufficiently large $kR$,*

$$\|\boldsymbol{P}_R - q\|_{L^2(D_R)} \le C_{\mathrm{cur}} (kR)^{2/3}, \tag{57}$$

*and*

$$\begin{aligned} \left| \frac{\sigma_{\mathrm{ext},R}}{R^2} - X_\infty \right| &\le C_{\mathrm{ext}} (kR)^{-1/3}, \\ \left| \frac{\sigma_{\mathrm{abs},R}}{R^2} - A_\infty \right| &\le C_{\mathrm{abs}} (kR)^{-1/3}, \\ \left| \frac{\sigma_{\mathrm{sca},R}}{R^2} - C_\infty \right| &\le C_{\mathrm{sca}} (kR)^{-1/3}. \end{aligned} \tag{58}$$

*At fixed $k$, (57) is equivalently $O(R^{2/3})$ in fixed length units. The rate is a valid upper bound; optimality of the exponent is not claimed. Constants need not remain bounded as $\delta \downarrow 0$ or $L \to \infty$.*

*Proof.* Use the cutoff from Lemma 3.7 and put $H_{R,w} = T(g_{R,w}q) - g_{R,w}T_0q$. The Fourier representation and its $H$-valued density justification are the same as in Lemma 3.6, with $R, w$ fixed during that justification. At low frequencies,

$$\|H^{\mathrm{low}}_{R,w}\|_{L^2(\mathbb{R}^2;H)} \le C\|q\|_H (R/w)^{1/2}. \tag{59}$$

At high frequencies, apply (46) with $N = 5$ and the last bound of (56). Inverse Fourier $L^1$-to-$L^\infty$ control gives

$$\begin{aligned} \sup_x \|H^{\mathrm{high}}_{R,w}(x,\cdot)\|_H &\le C\|q\|_H R w^{-4}, \\ \|H^{\mathrm{high}}_{R,w}\|_{L^2(D_R)} &\le C\|q\|_H R^2 w^{-4}. \end{aligned} \tag{60}$$

The second line uses $|S_R|^{1/2} = R$. In particular, no square of the grazing singularity is integrated.

The residual identity (49), uniform coercivity, and comparison of $g_{R,w}q$ with $q$ now yield

$$\|\boldsymbol{P}_R - q\|_{L^2(D_R)} \le C\left((Rw)^{1/2} + (R/w)^{1/2} + R^2w^{-4}\right), \tag{61}$$

where all factors involving the fixed $f, q, k, L, \delta, K$ have been included in $C$. To specify length units correctly, take the fixed reference length $\ell = 1/k$ and choose

$$w = \ell(R/\ell)^{1/3} = k^{-1}(kR)^{1/3}. \tag{62}$$

For sufficiently large $kR$, $w \le R/8$. If $\rho = kR$, the three terms in (61) have orders $k^{-1}\rho^{2/3}$, $\rho^{1/3}$, and $k^2\rho^{2/3}$, respectively. Since $k$ is fixed, this proves (57).

The incident, planar-current, and exact-current norms on $D_R$ are all $O(R)$. The extinction difference is therefore $O(R(kR)^{2/3})$, by its linear volume formula. The bounded quadratic form $W$ gives the same bound for absorption, as in (54). Division by $R^2$ gives $O((kR)^{-1/3})$, with the fixed factors of $k$ absorbed into the constants. Subtracting the exact extinction and absorption identities gives the full scattering error in (58). □

### 3.6 An exact planar pair with arbitrary fixed upper material caps

We next give the finite dielectric construction used before truncation. The tensors are ordinary local volume coefficients.

**Lemma 3.9** (Two exact dielectric layers)**.** *Fix $k > 0$, $P > 1$, $-1 < t < 1$, $\eta > 0$, and $h > 0$. There is a two-layer, nonmagnetic, reciprocal planar dielectric with $I_3 \preceq \operatorname{Re}\epsilon \preceq PI_3$ and $\operatorname{Im}\epsilon \succeq 0$ for which opposite $y$-polarized incidences obey*

$$A_+ = 0, \qquad A_- = (1-t^2)A_x, \qquad X_+ = X_- = 2(1-t), \qquad \frac{|A_+ - A_-|}{X_+} = \frac{1+t}{2}A_x. \tag{63}$$

*Here $0 < A_x < 1$ is the exact absorptance of a scalar slab of permittivity $1 + i\eta$ and thickness $h$. Taking $\eta_m = 1/m$, $kh_m = m^2$, and $t_m = 1 - 1/m$ gives a ratio tending to one, with finite coefficients and finite thickness for every integer $m \ge 2$.*

*Proof.* Choose once and for all a positive integer $j$ such that

$$p_j = \left(\frac{2j}{2j-1}\right)^2 \le P, \qquad d = \frac{(2j-1)\pi}{k}.$$

Such a $j$ exists because $p_j \downarrow 1$. Let $s = \sqrt{1-t^2}$ and

$$V = \begin{pmatrix} -t & s \\ s & t \end{pmatrix}, \qquad V^T = V, \qquad V^2 = I_2.$$

The first layer, on $0 < z < h$, is

$$\epsilon_A = \mathrm{diag}(1 + i\eta, 1, 1). \tag{64}$$

The second layer, on $h < z < h + d$, has zero transverse–longitudinal entries and

$$\epsilon_{W,\perp} = \frac{(p_j+1)I_2 - (p_j-1)V}{2}, \qquad \epsilon_{W,zz} = 1. \tag{65}$$

It is real symmetric, with transverse eigenvalues 1 and $p_j$. For the eigenvectors of $V$ with eigenvalues $+1$ and $-1$, its optical phases are respectively $(2j-1)\pi$ and $2j\pi$.

For a scalar slab of index $n$, continuity of the tangential electric and magnetic fields is equivalent here to continuity of $E$ and $E'$. Propagation by a distance $d$ acts on $(E, E'/k)^T$ as

$$\begin{pmatrix} \cos(nkd) & \sin(nkd)/n \\ -n\sin(nkd) & \cos(nkd) \end{pmatrix}.$$

At phase $m\pi$, this is $(-1)^m I_2$; between vacuum exteriors the reflection is zero and the boundary-plane transmission is $(-1)^m$. Thus the exact plate is reflectionless for both polarizations and has boundary-plane Jones transmission

$$\mathsf{H} = -V, \qquad e^{-ikd}\mathsf{H} = V. \tag{66}$$

This also proves the required vacuum-reference phase for every prescribed cap $P > 1$.

The absorber's $y$ polarization sees vacuum. For its $x$ polarization let $n = \sqrt{1+i\eta}$ with positive real and imaginary parts. Matching $E, E'$ at its two interfaces gives

$$r_0 = \frac{1-n}{1+n}, \qquad p = e^{inkh}, \qquad r_A = \frac{r_0(1-p^2)}{1-r_0^2p^2}, \qquad \tau_A = \frac{(1-r_0^2)p}{1-r_0^2p^2}, \qquad A_x = 1 - |r_A|^2 - |\tau_A|^2. \tag{67}$$

Since $\mathrm{Re}\, n > 0$ and $\mathrm{Im}\, n > 0$, $|r_0| < 1$ and $|p| < 1$. The denominator in (67) cannot vanish and $\tau_A \neq 0$, so $A_x < 1$. The exact flux identity

$$A_x = k\eta \int_0^h |E_x(z)|^2\, dz$$

gives $A_x > 0$: a zero integral would force $E_x$ and $E_x'$ to vanish inside, whereas the incident interface would require both $1 + r_A = 0$ and $1 - r_A = 0$.

For clarity, the full four-port scattering matrix, with boundary planes at $z = 0, h + d$ and channels ordered as $(Lx, Ly, Rx, Ry)$, is

$$\mathsf{S} = \begin{pmatrix} \mathsf{R}_A & \mathsf{D}_A\mathsf{H} \\ \mathsf{H}\mathsf{D}_A & \mathsf{H}\mathsf{R}_A\mathsf{H} \end{pmatrix}, \qquad \mathsf{R}_A = \mathrm{diag}(r_A, 0), \qquad \mathsf{D}_A = \mathrm{diag}(\tau_A, e^{ikh}). \tag{68}$$

There is no omitted multiple-reflection denominator because the plate has exactly zero reflection from either side. A $+z$ incident $y$ field passes through the absorber without loss; the plate then rotates it without absorption. A $-z$ incident $y$ field crosses the plate first, producing $\mathsf{H}e_y = -se_x - te_y$, and its absorber loss is $s^2A_x$. The co-polarized boundary-plane transmission in either direction is

$-te^{ikh}$. Dividing by the vacuum propagation $e^{ik(h+d)}$ gives exactly $t$, by (66). The planar extinction formula (40) now proves (63). All four output channels are retained, so polarization conversion is included in scattering.

Finally, for $\eta_m = 1/m$ and $kh_m = m^2$,

$$n_m \longrightarrow 1, \qquad r_{0,m} \longrightarrow 0, \qquad m \operatorname{Im} n_m = \frac{1}{2 \operatorname{Re} n_m} \longrightarrow \frac{1}{2}.$$

Hence $|p_m| = \exp[-m^2 \operatorname{Im} n_m] \to 0$. Equation (67) gives $r_{A,m} \to 0$, $\tau_{A,m} \to 0$, and $A_{x,m} \to 1$. With $t_m = 1 - 1/m$, the last ratio in (63) therefore tends to one. The real permittivity eigenvalues in both layers stay in $[1, P]$. □

**Lemma 3.10** (Positive-loss regularization at a fixed finite stack)**.** *Fix one finite stack in Lemma 3.9, with $-1 < t < 1$. Replace its two tensors by*

$$\epsilon_{A,\delta} = \epsilon_A + i\delta I_3, \qquad \epsilon_{W,\delta} = \epsilon_W + i\delta I_3. \tag{69}$$

*Its planar scattering amplitudes, power densities, and asymmetry ratio are continuous as $\delta \downarrow 0$. For every fixed $\delta > 0$, the susceptibility satisfies (14) for a finite $K$ and that positive loss floor $\delta$.*

*Proof.* This continuity statement is proved in the finite-dimensional one-dimensional differential equation, not by evaluating $\chi^{-1}$ at $\delta = 0$, where $\chi$ may be singular. The transverse fields obey

$$\frac{d}{dz}\begin{pmatrix} E_\perp \\ E'_\perp/k \end{pmatrix} = k \begin{pmatrix} 0 & I_2 \\ -\epsilon_\perp(\delta) & 0 \end{pmatrix} \begin{pmatrix} E_\perp \\ E'_\perp/k \end{pmatrix}.$$

At fixed finite layer thickness, each propagation matrix is a matrix exponential analytic in $\delta$. Their product, and the finite boundary-matching system for reflection and transmission, are therefore continuous at zero.

The matching matrix is invertible at zero. Indeed, a homogeneous outgoing solution would, by the passive one-dimensional flux identity, satisfy

$$|\text{left outgoing amplitude}|^2 + |\text{right outgoing amplitude}|^2 + k \int_0^{h+d} E_\perp^\dagger \operatorname{Im} \epsilon_\perp E_\perp \, dz = 0.$$

All three terms are nonnegative, so the exterior amplitudes vanish. Then $E_\perp = E'_\perp = 0$ at an interface, and uniqueness for the ordinary differential equation makes the field identically zero. Thus the matching system has trivial kernel and remains invertible for sufficiently small $\delta$. The scattering amplitudes and their power expressions are continuous. Since the unregularized common extinction is $2(1 - t) > 0$, the ratio is also continuous in a neighborhood of zero.

Adding $i\delta I_3$ makes $\operatorname{Im}(\epsilon - I_3) \succeq \delta I_3$ in each layer, without changing its real part or complex symmetry. The two finite tensors have a finite common norm bound. This proves the last assertion and permits application of Theorem 3.1 after $\delta$ has been fixed. □

### 3.7 The compact supremum and its order of limits

**Theorem 3.11** (Compact local nonmagnetic dielectric supremum)**.** *Fix $k > 0$, an upper real-permittivity cap $P > 1$, and an upper loss cap $Q > 0$. For every $\varepsilon > 0$, there is a compact three-dimensional scatterer made of two finite homogeneous dielectric layers, with $\mu = I_3$, vacuum exterior, and*

$$\epsilon^T = \epsilon, \qquad I_3 \preceq \operatorname{Re}\epsilon \preceq PI_3, \qquad 0 \prec \operatorname{Im}\epsilon \preceq QI_3 \quad \textit{inside both layers}, \tag{70}$$

*such that opposite $y$-polarized plane waves have common positive extinction $E$ and*

$$\Gamma = \frac{|\sigma^+_{\mathrm{abs}} - \sigma^-_{\mathrm{abs}}|}{E} > 1 - \varepsilon. \tag{71}$$

*Thus the unrestricted compact-class supremum of $\Gamma$ under these fixed upper caps is one. Every individual regular compact passive reciprocal scatterer with positive common extinction satisfies $\Gamma < 1$.*

*Proof.* It suffices to consider $0 < \varepsilon < 1$. First choose a finite member $(\eta, h, t)$ of Lemma 3.9 such that

$$\Gamma_{\mathrm{planar}} > 1 - \varepsilon/3, \qquad \eta < Q/2.$$

The same fixed integer $j$ realizes the prescribed cap $P$ for all members. Next use Lemma 3.10 to choose a fixed finite $0 < \delta < Q/2$ sufficiently small that

$$\Gamma_{\mathrm{planar},\delta} > 1 - 2\varepsilon/3.$$

The two loss eigenvalue sets are $\{\eta + \delta, \delta, \delta\}$ and $\{\delta, \delta, \delta\}$, so (70) holds. The selected regularized planar stack has fixed thickness $L = h + d$, fixed positive loss floor, and positive common extinction density $X_\delta$.

Truncate these two fixed layers to the side-$R$ square $(-R/2, R/2)^2$. Apply Theorem 3.1, or the quantitative estimate of Proposition 3.8, to each incident direction. Exact reciprocity gives equal extinction for every finite $R$. Since its area-normalized limit $X_\delta$ is positive, the ratio of compact cross sections converges:

$$\frac{|\sigma^+_{\mathrm{abs},R} - \sigma^-_{\mathrm{abs},R}|}{\sigma^+_{\mathrm{ext},R}} \longrightarrow \frac{|A^+_{\infty,\delta} - A^-_{\infty,\delta}|}{X_\delta} = \Gamma_{\mathrm{planar},\delta}.$$

Thus a sufficiently large but finite $R$ makes the ratio exceed $1 - \varepsilon$. For a fixed selected stack, the square estimate also gives a ratio error $O((kR)^{-1/3})$: once $\sigma^+_{\mathrm{ext},R}/R^2 \geq X_\delta/2$, ordinary quotient estimates apply. The constant can depend on $X_\delta^{-1}$ and on the fixed-stack constants.

The upper bound and its strictness follow from the complete optical theorem. For any such reciprocal pair with common extinction $E > 0$, passivity gives $0 \leq A_\pm \leq E$, hence $|A_+ - A_-|/E \leq 1$. If equality held, the more absorbing illumination would have absorption $E$ and therefore zero scattering. Its continuous far-field amplitude would have zero integrated squared norm and would vanish in every direction. The forward optical theorem (29) would then give $E = 0$, a contradiction. Together with the finite examples just constructed, this proves the asserted supremum and nonattainment. □

The limiting choices are ordered: first choose a finite planar member, then fix its positive regularization loss, and only then enlarge its aperture. Repeating this finite selection for a sequence

of tolerances is a diagonal construction, not an interchange of a lossless limit with an infinite-aperture limit. In particular, no estimate uniform in $\delta$, $L$, or the vanishing planar extinction has been used. Every coefficient and dimension is finite for every resulting compact body. The sequence has no common bound on thickness, footprint, diameter, volume, or integrated material loss; its positive loss floor may decrease from one member to the next. The theorem therefore establishes the supremum with fixed frequency and fixed upper material caps, without asserting a fixed-resource gap law or a bandwidth result.

# 4 Causal material interpretation and resource accounting

For each constructed layer, its Hermitian real and imaginary parts commute and have a common real orthogonal eigenbasis at the working frequency. A target eigenvalue $1 + a + ib$ with $a \geq 0$, $b > 0$ has a causal passive Lorentz extension. Choose any damping $\gamma > 0$, set

$$\Delta = \frac{a}{b}\gamma\omega, \qquad \omega_0^2 = \omega^2 + \Delta, \qquad F = \gamma\omega\,\frac{a^2 + b^2}{b} > 0,$$

and define, for a variable frequency $\zeta$,

$$\varepsilon_{\mathrm{eig}}(\zeta) = 1 + \frac{F}{\omega_0^2 - \zeta^2 - i\gamma\zeta}. \tag{72}$$

At $\zeta = \omega$ this is exactly $1+a+ib$. Its poles lie in the lower half-plane, its loss is nonnegative at positive real frequency, and its oscillator strength is positive. Independent principal-axis oscillators and real rotations give local reciprocal tensor dispersion. Thus the selected single-frequency coefficients do not require gain or a noncausal point response. No finite-bandwidth performance statement follows.

Passivity also has a direct time-domain proof. Each real principal-axis polarization obeys $\ddot{p} + \gamma\dot{p} + \omega_0^2 p = F\mathcal{E}$ and

$$\frac{d}{ds}\frac{\dot{p}^2 + \omega_0^2 p^2}{2F} = \mathcal{E}\dot{p} - \frac{\gamma}{F}\dot{p}^2,$$

where dots denote physical time derivatives. The stored oscillator energy and the damping dissipation are nonnegative. Summing this identity in the real orthogonal principal axes proves passivity of the tensor extension.

The compact examples need not have small *total* cross sections when $\Gamma$ approaches one. For each fixed planar member, $\sigma_{\mathrm{ext}} \sim R^2 X_\infty$. Although the selected planar extinction density can tend to zero, the aperture is chosen afterward and may grow much faster. The product can grow. Transferring the dipole conclusion "$\Gamma \to 1$ requires $\sigma_{\mathrm{ext}} \to 0$" to unrestricted compact objects would be incorrect.

The pointwise caps $P, Q$ remain fixed. Total volume, footprint, and integrated loss can grow, and the coercive loss floor can decrease. The proof does not optimize that cost. The square-aperture error bound controls a fixed stack; its constants depend on the stack, including its thickness and loss floor, and are not uniform along the final three-parameter sequence.

# 5 Numerical tests and independent verification

## 5.1 Radiation-consistent dipolar checks

The exact six-coordinate model was tested with independent spherical quadrature, direct radiation removal and reapplication, and 2,000 randomly generated reciprocal contractions. These are checks

on the stated dipole model; they are not a local dielectric completion theorem. The largest absolute full-sphere radiation discrepancy was $7.11 \times 10^{-15}$ in units of $k^{-2}$; the largest Frobenius-norm error in recovering the dimensionless dressed polarizability from its bare response was $5.05 \times 10^{-16}$. No random sample exceeded the proved envelope. The final equality-family sample, $t = 0.99999999$, gave $\Gamma = 0.999999995$. Tiny negative computed null-eigenvalue or zero-absorption values at floating-point roundoff are retained in the JSON.

The exploratory constrained SciPy SLSQP runs are also retained: of 16 starts, 8 reported failure or an iteration limit, and 2 successful exits were visibly below the proved optimum. Optimizer success is not a global optimality certificate. The sharp envelope and the exact equality family have algebraic proofs independent of these attempts.

### 5.2 Finite-aperture full-vector experiments

We implemented a nonperiodic three-dimensional retarded tensor discrete-dipole approximation (DDA), using the established radiation-reaction formulation [13, 14]. A cubic cell of side $\Delta$ has

$$\alpha_0 = 3\Delta^3(\epsilon - I)(\epsilon + 2I)^{-1}, \qquad \alpha^{-1} = \alpha_0^{-1} - \frac{ik^3}{6\pi} I.$$

Every cell is reciprocal and passive. The interaction contains the full electric dyadic Green tensor with its retarded near, intermediate, and far terms. All three electric components are solved; their retarded current distribution generates the full magnetic field. FFT padding is at least $2n - 1$ in each dimension and implements ordinary finite-domain convolution, not transverse periodicity. The physical object is the entire finite square prism, including its vertical edges and corners.

For solved cell moments $p_i$ and off-diagonal Green interaction $G$, the scattering calculation uses the exact complete-sphere radiation Gram form

$$\sigma_{\mathrm{sca}} = k \left[ \frac{k^3}{6\pi} \sum_i |p_i|^2 + \Im(p^\dagger G p) \right].$$

It is not calculated by collecting only transmitted and reflected specular waves. Absorption uses $k \sum_i p_i^\dagger [-\Im \alpha_{0,i}^{-1}] p_i$; extinction uses $k \Im \sum_i E_{\mathrm{inc},i}^\dagger p_i$. An independent full-sphere angular quadrature checks the same radiation.

The exact planar comparison uses $4 \times 4$ homogeneous Maxwell matrix exponentials and boundary matching, with the vacuum reference phase retained. The supplied tensors have zero transverse–longitudinal coupling. The code explicitly rejects other tensors because a general planar tensor requires longitudinal-field elimination before constructing that generator.

Parameter set A has $t = 0.7$, $\eta = 0.8$, $\delta = 0.002$, $h = \lambda$, and plate depth $\lambda/2$, with real plate eigenvalues $1, 4$. Parameter set B has $t = 0.9$, $\eta = 0.3$, $\delta = 5 \times 10^{-5}$, $h = 2\lambda$, and the same plate depth. The tabulated grid parameter is $\lambda/\Delta$.

Set A's exact planar ratio is 0.782117503. Set B's exact planar ratio is 0.918059609. The solver reached its iteration limit in 2 cases. It used at most 30 restarted GMRES cycles of 35 steps for each incidence; its requested preconditioned relative tolerance was $2 \times 10^{-8}$. The complete residual histories and exit codes are preserved. These exits identify a numerical linear-solver limit, not a failure of the physical construction or of the localization proof. At fixed $R = 2\lambda$, changing the mesh from $\lambda/12$ to $\lambda/24$ changes $\Gamma$ from 0.639717 to 0.660781. This measures discretization sensitivity; it does not supply a certified continuum error bar or establish a fitted convergence order. Among converged production cases, the largest optical-theorem relative discrepancy is $1.03 \times 10^{-8}$ and the largest opposite-extinction relative discrepancy is $1.17 \times 10^{-8}$. For the separate $R = 0.75\lambda$ check, independent angular integration of the complete radiated field agrees with the radiation Gram form within $5.11 \times 10^{-15} \lambda^2$.

| Set | $R/\lambda$ | $\lambda/\Delta$ | Cells | $\Gamma$ | Residual | Status |
|---|---|---|---|---|---|---|
| A | 1 | 16 | 6,144 | 0.558592 | $1.5 \times 10^{-8}$ | converged |
| A | 2 | 16 | 24,576 | 0.653536 | $1.9 \times 10^{-8}$ | converged |
| A | 4 | 16 | 98,304 | 0.715552 | $5.2 \times 10^{-9}$ | converged |
| A | 6 | 16 | 221,184 | $0.731455^{\dagger}$ | $1.4 \times 10^{-7}$ | limit |
| A | 2 | 12 | 10,368 | 0.639717 | $1.9 \times 10^{-8}$ | converged |
| A | 2 | 24 | 82,944 | 0.660781 | $1.4 \times 10^{-8}$ | converged |
| B | 4 | 16 | 163,840 | 0.826085 | $2.6 \times 10^{-9}$ | converged |
| B | 8 | 12 | 276,480 | $0.861586^{\dagger}$ | $2.0 \times 10^{-5}$ | limit |

Table 1: Actual finite-system calculations. The residual is the norm of the unpreconditioned Maxwell dipole equation divided by the incident-field norm. A dagger marks a solver that reached its iteration limit and is retained as a diagnostic, not counted as a converged calculation.

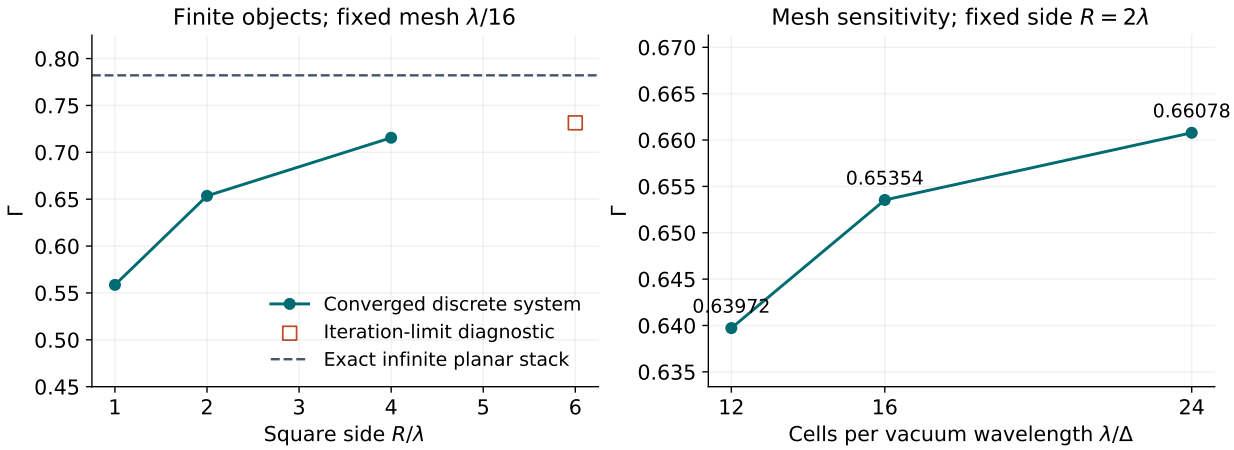



Figure 1: Finite-aperture and mesh diagnostics for parameter set A. The dashed line is the exact planar value. The finite-mesh data are not assigned continuum error bars, and the analytical aperture exponent is not fitted to these curves. Hollow squares denote iteration-limit diagnostics.

### 5.3 Exact infinite-planar limiting cases

An additional exact scattering calculation tests the regularized planar construction at $\eta = 1/m$, $kh = m^2$, $t = 1 - 1/m$, and the positive sampled regularization $\delta = m^{-6}$. It composes the two polarization scattering matrices using only decaying slab propagation factors. This avoids exponentially growing transfer-matrix entries at large optical thickness. These data are *infinite-planar power densities*, not compact-body cross sections.

| $m$ | $kh$ | $\delta$ | Exact planar $\Gamma$ ($P = 4$) |
|---|---|---|---|
| 4 | 16 | $2.4 \times 10^{-4}$ | 0.845238489 |
| 10 | 100 | $1.0 \times 10^{-6}$ | 0.948826698 |
| 20 | 400 | $1.6 \times 10^{-8}$ | 0.974783327 |
| 50 | 2500 | $6.4 \times 10^{-11}$ | 0.989971210 |
| 100 | 10000 | $1.0 \times 10^{-12}$ | 0.994993279 |

The same $m = 100$ sample with upper real cap $P = 1.01$ uses $j = 101$ and gives $\Gamma = 0.994993247$. All sampled maximum losses are below the common cap $Q = 0.3$. At three moderate-thickness comparison points, the stable scattering calculation and the independent $4 \times 4$ matrix-exponential calculation agree in all absorption, extinction, and scattering densities within $1.12 \times 10^{-15}$. The complete four-port passivity defect and reciprocal matrix symmetry are also checked. These finite samples do not replace the analytic slab limit or establish uniform aperture estimates for $\delta = m^{-6}$.

### 5.4 Independent audits and their limits

Separate AI-agent mathematical reviews checked the Green operator, coercivity, grazing-frequency estimate, square cutoff, exact plate phases, loss regularization, and ordered limits. A further transcription review rederived the complete extinction reciprocity and the passage from smooth longitudinal profiles to $L^2$ profiles. These are ordinary mathematical reviews, not a proof-assistant certificate or external peer review. Their written findings are included separately in the archive.

An independently coded solver audit assembles the dense nonperiodic dyadic Green matrix and its complete radiation Gram matrix, then compares them with the FFT implementation. It includes genuinely nonnormal complex symmetric tensors with positive material loss, as well as FFT dimensions larger than the minimum nonperiodic padding. Its recorded maximum errors are:

| Check | Maximum relative discrepancy |
|---|---|
| Dense versus FFT interaction | $6.76 \times 10^{-16}$ |
| Full-sphere radiation Gram | $2.85 \times 10^{-15}$ |
| Optical-theorem energy balance | $4.78 \times 10^{-14}$ |
| Dense versus FFT solutions | $8.75 \times 10^{-14}$ |
| Opposite-incidence extinction | $6.31 \times 10^{-16}$ |

The DDA is a finite Maxwell discretization with exact radiation consistency for its point moments. It is not the exact local continuum body at a finite mesh spacing. The compact-continuum conclusion rests on the analytic proof in Section 3; no numerical agreement or optimizer status is used to discharge an unproved Maxwell realization step.

## 6 Prior results and the novelty boundary

| Closest source | Exact relevance and difference of scope |
|---|---|
| Sounas–Alù (2014) [2] | Extinction reciprocity and distinct absorption/scattering for reciprocal objects are established. Neither their existence nor that symmetry is a new contribution here. |
| Sersic et al. (2011) [3] | The full magnetoelectric quadratic optical theorem and inverse-polarizability radiation dressing are established. The later orthogonal-eigenvector reduction in that work does not cover arbitrary nonnormal matrices. |
| Mann–Alù (2023) [4] | The official conference abstract explicitly states maximal absorption for one incident direction and zero absorption for the opposite direction in reciprocal dipolar particles. Its full equations were not retrieved, so equivalence to the normalized optimum cannot be ruled out. No novelty claim is made for the dipole equality family. |
| Andersen–Frandsen (2005) [9] | The institutional abstract explicitly states that receiving-antenna absorption efficiency $\sigma_{\mathrm{abs}}/\sigma_{\mathrm{ext}}$ can approach one, with a 93% endfire-array example. That single-incidence objective does not by itself establish negligible absorption under the opposite illumination; the full-text comparison remains incomplete. |
| Hopkins et al. (2015) [5] | Reciprocal finite-particle absorption asymmetry and its nonorthogonal-mode mechanism are established. The present compact supremum is not inferred from a numerical example of that phenomenon. |
| Gao–Yang–Kattawar (2013) [6] | Infinite-plate extinction formulas and ADDA evidence of finite-plate convergence are established. The publisher text retrieved here describes numerical convergence. Our stated theorem instead proves all three cross-section limits for strictly absorbing tensor multilayers, with an explicit square-aperture error estimate. Full-text priority comparison is still incomplete. |
| Windowed Green functions [7, 8] | Smooth-window treatment of unbounded interfaces is established. These methods retain the infinite physical background; numerical truncation of an integral is not automatically truncation of the material body. Our trial window is used to estimate the exact sharp-boundary finite object. |
| 2017/2022/2026 bounds [17, 18, 19] | Time-reversal field bounds, reciprocal reflection feasibility, and coupled-mode metasurface optima retain their specified assumptions. They do not by themselves give the finite-aperture Maxwell completion used here. |

The search covers accessible primary literature through 10 October 2026. The exact compact localization proof and its normalized-supremum application have passed independent mathematical review, but their *priority is not certified*. In particular, the missing full Mann–Alù contribution and the incomplete full-text comparison with Gao et al. prevent an unconditional NEW label. Mathematical proof and a priority claim are separate judgments.

# 7 Implications for OQ7 and remaining obligations

For the question stated in Theorem 1.1, there is a definite answer: no uniform number $c < 1$ bounds $\Gamma$ across all compact local reciprocal dielectrics of unrestricted size, even with fixed positive frequency and fixed upper real-permittivity and loss caps. Equality is approached by finite objects and is unattainable at positive extinction. This closes the normalized resource-free compact-body supremum problem posed for this phase.

It does not establish a complete catalogue of the resource-dependent bounds suggested by the broader original OQ7. The following remain open here:

1. Sharp laws at fixed diameter, volume, thickness, aperture, integrated loss, or a prescribed positive tensor-loss floor.

2. A sharp cost of approaching $\Gamma = 1$. The proved aperture rate is an upper estimate for each fixed regularized stack, not an optimal global resource law.

3. The supremum at prescribed nonzero total extinction, particularly with a bounded footprint, and finite-bandwidth dispersion constraints.

4. An exact local finite-particle realization theorem for every radiation-consistent dipolar equality tensor, including uniform control of resonant small-sphere asymptotics. This is unnecessary for the large-aperture counterexample, but remains an independent realization question.

5. Certified finite-mesh continuum error bars for the particular DDA geometries, and a full prior-theorem comparison sufficient for priority.

The general normalized supremum, the optical theorem, and the all-angle localization limit have ordinary mathematical proofs in this report. No Lean formalization was attempted, no prior Lean source was changed, and no existing formal theorem is cited as proving these new analytic steps.

# A Reproducibility and preservation

The archive contains the exact proof sources, separate independent reviews, all numerical programs and JSON outputs, and the failed exploratory dipole optimizations. It also includes the unchanged preceding manuscript and its complete source archive, together with a before/after hash audit of the preceding mathematical and numerical sources. It does not overwrite the earlier research version.

Use Python 3.12, NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. From the new research root, the numerical entry points are

```
python3 numerics/dipole_checks.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=2 \
  python3 numerics/aperture_sweep.py
python3 numerics/aperture_independent_check.py
python3 numerics/planar_limit_checks.py
python3 numerics/make_figures.py
```

The aperture sweep saves each completed calculation before beginning the next and reuses existing checkpoints. The README explains how to run a fresh reproduction in a separate directory. No solver output is used as a replacement for the analytic localization proof.

Build this document with

```
cd report
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  oq7_compact_cross_sections.tex
```

The theorem statements and their complete hypotheses, rather than an informal summary or a solver success flag, determine the proved scope.

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